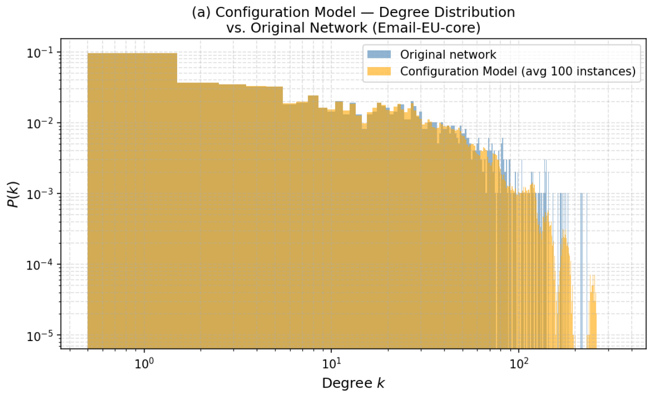
Network Science – Assignment 3 Report, Group 18

1. Degree Distribution Under Configuration Model and Edge-Swapping

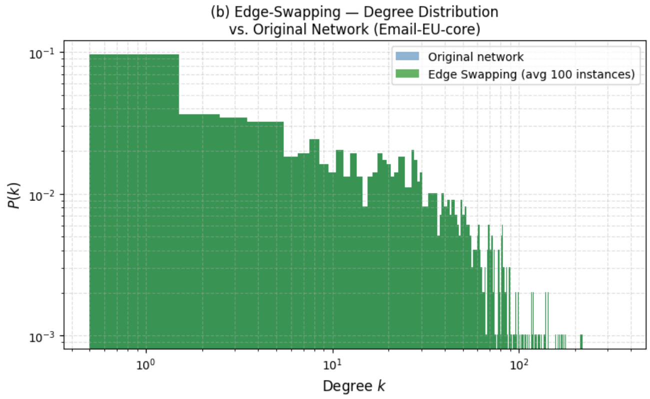
Dataset. Email-EU-core network (SNAP): a communication network from a European research institution loaded as an undirected simple graph. **986 nodes, 16,064 edges**, mean degree 32.6, max degree 345.

Configuration Model. Each node receives stubs equal to its degree; stubs are paired uniformly at random and self-loops / multi-edges are discarded. 100 instances were generated and degree distributions averaged.

Edge-Swapping. Starting from the original graph, two edges are selected at random and their endpoints swapped if the swap creates no self-loops or duplicate edges. $Q = 5$ accepted swaps per edge per instance (100 instances). The degree sequence is preserved exactly in every instance.



(a) Configuration Model (log-log).



(b) Edge-Swapping (log-log).

Figure 1: Averaged degree distributions (100 instances) vs. original.

Observations. Both methods reproduce the original degree distribution well. The Configuration Model shows a small high-degree deficit because high-degree nodes produce more stubs and are more likely to generate discarded self-loops or multi-edges. Edge-Swapping matches the original exactly since node degrees never change during a swap. Both serve as degree-preserving null models for studying higher-order structural properties.

2. Degree Correlations (k_{nn} vs k)

Method. For each node i, the average nearest-neighbour degree is $k_{nn}(i) = \frac{1}{k_i}\sum_{j\in\mathcal{N}(i)} k_j$. Values are binned by k and averaged to form $k_{nn}(k)$. A random-graph baseline (100 Configuration Model instances) provides the uncorrelated reference.

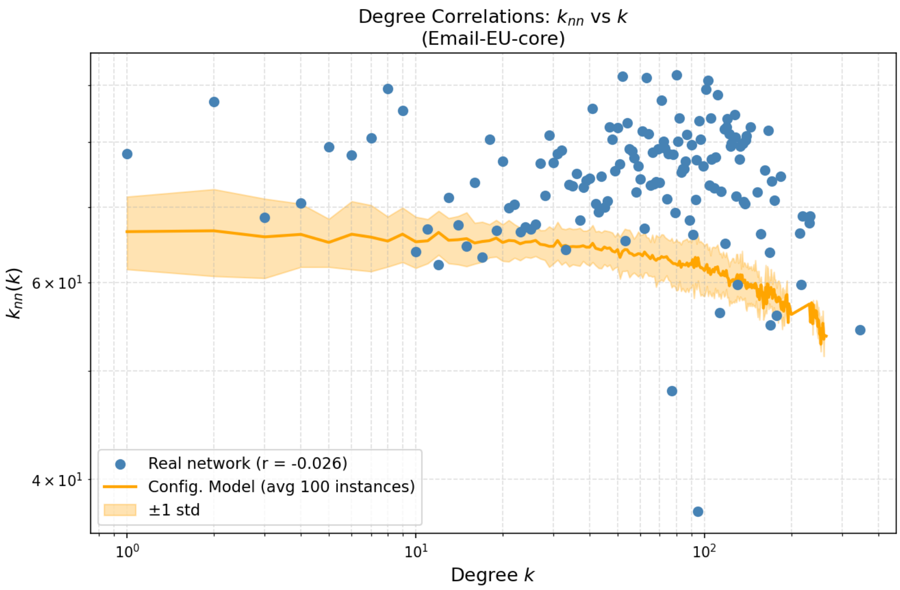


Figure 2: k_{nn} vs k for the real network and the random-graph baseline (±1 std). Pearson $r = -0.0257$.

Observations. Pearson assortativity $r = -0.0257$, indicating a nearly uncorrelated network. The $k_{nn}(k)$ curve is mostly flat and tracks the random baseline closely, showing that most of the correlation structure is already explained by the degree sequence alone.

3. Residue Interaction Graph (RIG) and Long-Range Interaction Network (LIN)

Selected proteins.

PDB ID	Protein	Organism	Structure	Residues
1UBQ	Ubiquitin	Human	Mixed α/β	76
1HRC	Cytochrome C	Horse heart	Mostly α	104
1FKB	FKBP12	Human	β-sheet rich	107
1APS	Acylphosphatase	Human	Mixed α/β	98
1CSP	Cold Shock Prot. B	*B. subtilis*	β-barrel	67

Construction. RIG: edge between residues i and j if $C\alpha$–$C\alpha$ distance < 8 Å and $|i-j| \geq 2$. **LIN:** same distance cutoff, but $|i-j| > 12$ (long-range only).

Table 1: Network properties for RIG and LIN models.

	RIG				LIN			
Protein	Edges	$\langle k \rangle$	L	C	Edges	$\langle k \rangle$	L	C
1UBQ	251	6.61	3.16	0.326	90	2.37	5.41	0.000
1HRC	385	7.40	3.49	0.336	115	2.21	6.43	0.008
1FKB	433	8.09	3.35	0.346	212	3.96	4.27	0.025
1APS	392	8.00	3.16	0.327	205	4.18	4.23	0.030
1CSP	242	7.22	2.77	0.373	102	3.05	3.66	0.000

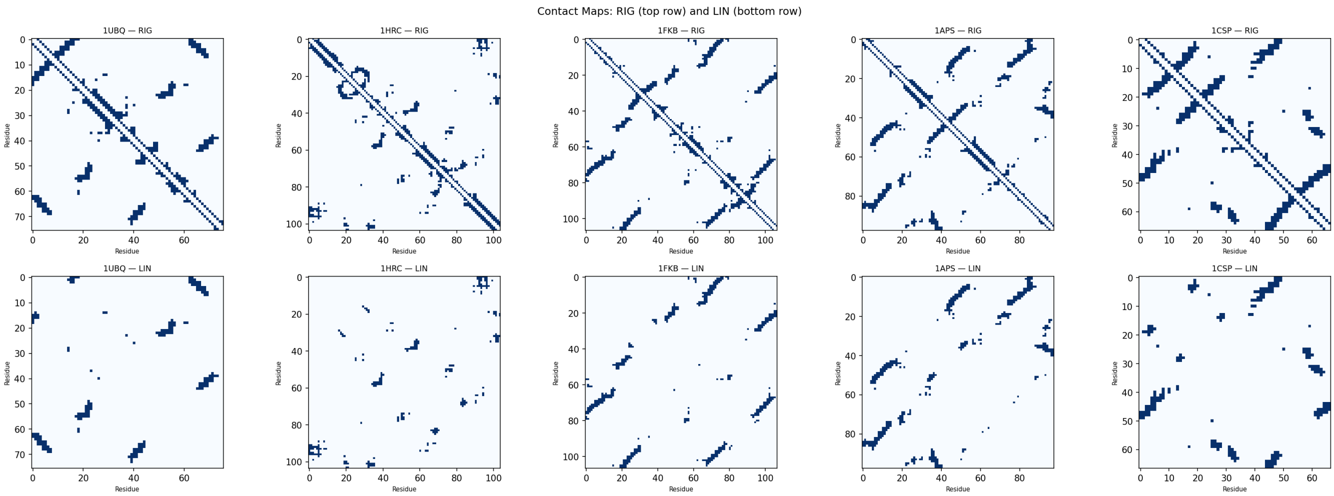


Figure 3: Contact maps: RIG (top) and LIN (bottom) for all five proteins.

Observations. All RIG networks show small-world character: $L \approx 2.8$–3.5 and $C \approx 0.33$–0.37, reflecting the dense local packing of protein structures. LIN networks are much sparser; β-sheet proteins (1FKB, 1APS) retain more LIN edges because β-sheets bring sequentially distant residues into 3D proximity. LIN clustering is near zero ($C \approx 0$–0.03) as long-range contacts rarely form triangles. Proteins with a high LIN/RIG edge ratio (1FKB, 1CSP) are expected to fold more slowly because forming long-range contacts requires a larger conformational search.

4. Bartoli's Model of Protein Structure

Model procedure. (i) Fill the first two off-diagonals of the adjacency matrix (backbone). (ii) Sample a pair (i, j) with $P \propto 1/|i-j|$. (iii) Set all 9 entries of $\{i \pm 1\} \times \{j \pm 1\}$ to 1. (iv) Repeat until edge count reaches the target (real RIG edge count). 100 instances were run per protein.

Table 2: L and C for real structures, standard Bartoli, and modified Bartoli (mean $\pm$ std, 100 instances).

	Real RIG		Real LIN		Bartoli		Mod. Bartoli	
Protein	L	C	L	C	L	C	L	C
1UBQ	3.16	0.326	5.42	0.000	4.07±0.49	0.578	5.67±0.79	0.603
1HRC	3.49	0.335	6.43	0.008	4.01±0.34	0.564	6.46±0.96	0.604
1FKB	3.35	0.346	4.27	0.025	3.65±0.27	0.561	5.97±0.86	0.605
1APS	3.16	0.327	4.23	0.030	3.52±0.22	0.563	5.55±0.83	0.604
1CSP	2.77	0.373	3.66	0.000	3.52±0.49	0.586	4.48±0.59	0.603

Part (a): Comparison with real RIG and LIN. Observations. Standard Bartoli produces L values moderately above the real RIG (≈ 3.5–4.1 vs. real ≈ 2.8–3.5) and significantly over-estimates clustering ($C \approx 0.56$–0.59 vs. real ≈ 0.33–0.37). The over-clustering arises because assigning 9 contacts at once in step (iii) creates artificial triangles at a density not found in real proteins.

Modified step (ii): We tested $P(i,j) \propto \frac{1}{|i-j|} \exp\left(-\frac{(|i-j|-8)^2}{72}\right)$, peaking at $|i-j| \approx 8$ to mimic secondary-structure contact distances. This *increased* L further from the real RIG rather than reducing it: concentrating contacts at medium range creates a more chain-like structure with longer paths. Clustering remained similarly elevated. A meaningful improvement would require incorporating residue-type and hydrophobicity information rather than sequence distance alone.

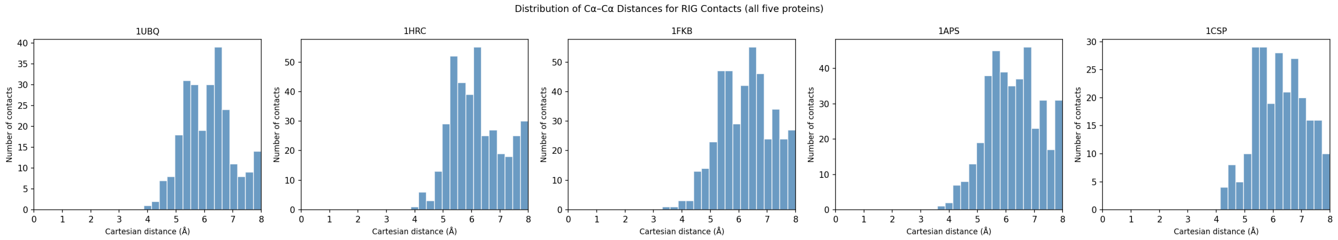


Figure 4: Distribution of $C\alpha$–$C\alpha$ distances across all RIG contacts for each protein (8 Å cutoff).

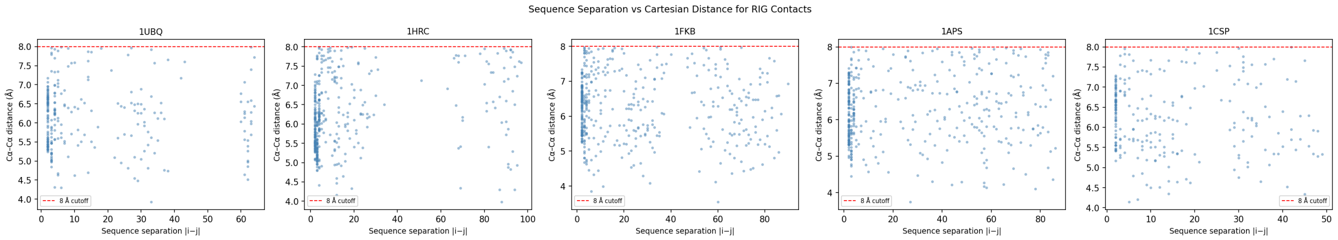


Figure 5: Sequence separation $|i - j|$ vs. 3D distance for all RIG contacts. Red dashed line marks the 8 Å cutoff.

Part (b): Contacts vs Cartesian distance. Observations. RIG contacts peak in the 6–7 Å range, consistent with the geometry of α-helices and adjacent β-strands. Short-range sequence contacts ($|i - j| < 5$) are tightly clustered at 5–7 Å due to backbone constraints. Long-range contacts spread more broadly within the 8 Å cutoff, as they are formed purely by the tertiary fold. This bimodal character suggests that a more accurate generative model should use a two-component probability: one peak for local secondary-structure contacts and a smaller one for long-range tertiary contacts. The single-peaked distributions in both Bartoli variants fail to capture this structure.